EHM(X): 制約生成と solver の合成

型付け・算法・健全性・主要性・条件付き完全性

1 構文とパラメータ

$$
\begin{aligned}
&e ::= i \mid n \mid b \mid \lambda.e \mid e\,e \\
&\quad \mid \mathbf{let}\ e\ \mathbf{in}\ e \mid \mathbf{perform}\ o\ e \mid \mathbf{handle}_o(e, e_r, e_o), \\
&v ::= i \mid n \mid b \mid \lambda.e, \qquad i, o \in \mathbb{N}, \\
&\epsilon = (L, M) \in \mathcal{P}_f(\mathbb{N})^2, \quad \langle\epsilon\rangle_\rho = L \cup \bigcup_{\mu \in M} \rho_E(\mu), \\
&A = (A_T, A_E) \in \mathcal{P}_f(\mathbb{N})^2, \\
&\rho = (\rho_T : \mathbb{N} \to V, \rho_E : \mathbb{N} \to \mathcal{P}_f(\mathbb{N})).
\end{aligned}
$$

項変数は de Bruijn 添字. 操作節：$0 = k,\ 1 = x$.

$$
\begin{gathered}
T : \text{型式}, \quad V \neq \varnothing, \quad \alpha_i,\ \mathrm{Int},\ \mathrm{Bool},\ t \xrightarrow{\epsilon} u \in T, \quad \leq \subseteq V^2, \\
a \leq a, \qquad (a \leq b \wedge b \leq c) \Rightarrow a \leq c, \qquad (a \leq b \wedge b \leq a) \Rightarrow a = b, \\
a' \leq a \wedge b \leq b' \Rightarrow a \xrightarrow{E} b \leq a' \xrightarrow{E} b'.
\end{gathered}
$$

T の構成子の網羅性・単射性は仮定しない.

$$
\begin{aligned}
\langle\alpha_i\rangle_\rho &= \rho_T(i), & \langle\mathrm{lit}(b)\rangle_\rho &= \mathrm{groundLit}(b), \\
\langle a \xrightarrow{\epsilon} b\rangle_\rho &= \mathrm{groundArrow}(\langle a\rangle_\rho, \langle\epsilon\rangle_\rho, \langle b\rangle_\rho), & & \\
\langle\mathrm{rename}(f, g, t)\rangle_\rho &= \langle t\rangle_{(\rho_T \circ f, \rho_E \circ g)}, & \theta\alpha_i &= \theta_T(i).
\end{aligned}
$$

$$
\frac{\forall i.\ A \cap \mathrm{supp}(\theta_T(i)) = \varnothing \quad \forall j.\ A_E \cap \mathrm{FV}_E(\theta_E(j)) = \varnothing}{A \cap \mathrm{supp}(\theta t) = \varnothing}
$$

$$
\begin{aligned}
\theta^*\rho &= (i \mapsto \langle\theta_T(i)\rangle_\rho, j \mapsto \langle\theta_E(j)\rangle_\rho), & \mathrm{dom}\,\theta &: \text{有限}, \\
\langle\theta t\rangle_\rho &= \langle t\rangle_{\theta^*\rho}, & \rho \models \theta C &\iff \theta^*\rho \models C, \\
\rho =_{\mathrm{supp}\,t} \nu &\Rightarrow \langle t\rangle_\rho = \langle t\rangle_\nu, & \rho =_{\mathrm{supp}\,C} \nu &\Rightarrow (\rho \models C \iff \nu \models C), \\
\theta|_{\mathrm{supp}\,t} = \mathrm{id} &\Rightarrow \theta t = t, & \mathrm{supp}(\alpha_i) &= (\{i\}, \varnothing).
\end{aligned}
$$

$$
\begin{aligned}
\rho \models \exists A.C &\iff \exists \nu.\ \nu =_{\overline{A}} \rho \wedge \nu \models C, \\
\rho \models \forall B.\exists J.C &\iff \forall \nu =_{\overline{B}} \rho.\ \exists \omega =_{\overline{J}} \nu.\ \omega \models C, \\
C \models_X D &\iff \forall \rho.\ (\rho \models C \Rightarrow \rho \models D), \\
C \equiv_X D &\iff C \models_X D \wedge D \models_X C, & \mathrm{Sat}(C) &\iff \exists \rho.\ \rho \models C, \\
i \in \mathrm{FV}_T(C) &\iff \exists \{i\}_T.C \not\equiv_X C, & \mathrm{FV}(C) &\subseteq \mathrm{supp}\,C.
\end{aligned}
$$

$C, D \in \mathcal{C}_X$. $\top, \wedge, =, \leq, \subseteq, o \notin \epsilon$ は各意味値で解釈する.

2 宣言的型付け

$$\sigma ::= t \mid \forall A.D \Rightarrow t, \qquad \Gamma ::= [] \mid \sigma :: \Gamma, \qquad \mathrm{supp}(\forall A.D \Rightarrow t) = (\mathrm{supp}\, D \cup \mathrm{supp}\, t) \setminus A.$$

C が共通の前提では省略する．

$$\frac{\Gamma[i] = \sigma}{\Gamma \vdash i : \sigma!\varnothing} \qquad \frac{}{\Gamma \vdash n : \mathrm{Int}!\varnothing} \qquad \frac{}{\Gamma \vdash b : \mathrm{Bool}!\varnothing}$$

$$\frac{a :: \Gamma \vdash e : b!\eta}{\Gamma \vdash \lambda.e : (a \xrightarrow{\eta} b)!\varnothing} \qquad \frac{\Gamma \vdash f : (a \xrightarrow{\eta} b)!\epsilon_f \quad \Gamma \vdash x : a!\epsilon_x}{\Gamma \vdash f\, x : b!(\epsilon_f \cup \epsilon_x \cup \eta)}$$

$$\frac{\mathrm{Value}(v) \quad \Gamma \vdash v : \sigma!\varnothing \quad \sigma :: \Gamma \vdash e : t!\epsilon}{\Gamma \vdash \mathbf{let}\ v\ \mathbf{in}\ e : t!\epsilon}$$

$$\frac{\neg\, \mathrm{Value}(u) \quad \Gamma \vdash u : a!\eta \quad a :: \Gamma \vdash e : t!\epsilon}{\Gamma \vdash \mathbf{let}\ u\ \mathbf{in}\ e : t!(\eta \cup \epsilon)}$$

$$\frac{C; \Gamma \vdash e : a!\epsilon \quad C \models_X a \le b}{C; \Gamma \vdash e : b!\epsilon} \qquad \frac{C; \Gamma \vdash e : \sigma!\epsilon \quad C \models_X \epsilon = \eta}{C; \Gamma \vdash e : \sigma!\eta}$$

$$\frac{\mathrm{Value}(v) \quad A \cap \mathrm{FV}(C, \Gamma) = \varnothing \quad C \wedge D; \Gamma \vdash v : t!\varnothing}{C \wedge \exists A.D; \Gamma \vdash v : (\forall A.D \Rightarrow t)!\varnothing}$$

$$\frac{C; \Gamma \vdash e : (\forall A.D \Rightarrow t)!\epsilon \quad \mathrm{dom}\,\theta \subseteq A \quad C \models_X \theta D}{C; \Gamma \vdash e : \theta t!\epsilon}$$

$$\frac{C; \Gamma \vdash e : \sigma!\epsilon \quad A \cap \mathrm{FV}(\Gamma, \sigma, \epsilon) = \varnothing}{\exists A.C; \Gamma \vdash e : \sigma!\epsilon} \qquad \frac{C; \Gamma \vdash e : \sigma!\epsilon \quad D \models_X C}{D; \Gamma \vdash e : \sigma!\epsilon}$$

$$\frac{C; \Gamma \vdash e : \sigma!\epsilon \quad \sigma \equiv_\alpha \sigma'}{C; \Gamma \vdash e : \sigma'!\epsilon}$$

$$\begin{aligned}
\mathrm{Adm}_S(C) &\iff \exists R.\ S(R) \wedge C \equiv_X R, \\
\mathrm{Adm}_S(t) &\iff \top, \qquad\qquad \mathrm{Adm}_S(\forall A.D \Rightarrow t) \iff \mathrm{Adm}_S(D), \\
\mathrm{Adm}_S(\Gamma) &\iff \forall \sigma \in \Gamma.\ \mathrm{Adm}_S(\sigma), \\
\mathrm{Typing}_S(C, \Gamma, e, \sigma, \epsilon) &\iff \exists \mathcal{D} : (C; \Gamma \vdash e : \sigma!\epsilon).\ \mathrm{Valid}_S(\mathcal{D}).
\end{aligned}$$

Valid_S は導出の全節点を検査する．以下 $S = \mathrm{Sat}$.

$$\frac{\mathcal{D} : (C; \Gamma \vdash e : \sigma!\epsilon) \quad \mathrm{Sat}(C) \quad \mathrm{Adm}_{\mathrm{Sat}}(\Gamma)}{\mathrm{Valid}_{\mathrm{Sat}}(\mathcal{D})}$$

3 操作・handler・環境

$$\Sigma(o) = (B, a, b), \qquad \mathrm{Closed}(\Sigma) \iff \forall o, (B, a, b) = \Sigma(o).\ \mathrm{supp}(a, b) \subseteq B.$$

$$\frac{\Sigma(o) = (B, a, b) \quad \mathrm{Closed}(B, a, b) \quad \mathrm{dom}\,\theta \subseteq B \quad C; \Gamma \vdash u : \theta a ! \eta}{C; \Gamma \vdash \mathbf{perform}\ o\ u : \theta b ! (\{o\} \cup \eta)}$$

$$\mathrm{Split}_o(\eta, \delta) = \delta \subseteq \eta \wedge \eta \subseteq \{o\} \cup \delta \wedge o \notin \delta.$$

操作宣言の新鮮な改名を (B, a, b) とする.

$$\frac{\begin{array}{c} C; \Gamma \vdash e : t_i ! \epsilon_i \qquad C; t_i :: \Gamma \vdash e_r : t ! \epsilon_r \\ D; (b \xrightarrow{\chi} t) :: a :: \Gamma \vdash e_o : t_o ! \epsilon_o \\ C \models_X \forall B. \exists J. (D \wedge t_o \leq t \wedge \epsilon_o \subseteq \chi) \\ C \models_X \mathrm{Split}_o(\epsilon_i, \delta) \wedge \delta \subseteq \chi \wedge \epsilon_r \subseteq \chi \\ \mathrm{Scope}(B, J; C, \Gamma, t_i, t, a, b, \epsilon_i, \epsilon_r, \delta, \chi) \end{array}}{C; \Gamma \vdash \mathbf{handle}_o(e, e_r, e_o) : t ! \chi}$$

$$\mathrm{Scope} \iff \begin{cases} B \cap \mathrm{FV}(\Gamma) = \varnothing, \\ B \cap \mathrm{supp}(t_i, t) = \varnothing, \\ B_E \cap \mathrm{FV}_E(\epsilon_i, \epsilon_r, \delta, \chi) = \varnothing, \\ J \cap \big(\mathrm{supp}\,\Gamma \cup \mathrm{supp}(t_i, t, a, b) \cup B\big) = \varnothing, \\ J \cap \mathrm{FV}(C) = \varnothing, \\ J_E \cap \mathrm{FV}_E(\epsilon_i, \epsilon_r, \delta, \chi) = \varnothing. \end{cases}$$

$$\frac{C \models_X a \leq b}{C \vdash a \preceq b} \qquad \frac{\mathrm{dom}\,\theta \subseteq A \quad C \models_X \theta D \quad C \vdash \theta t \preceq \sigma}{C \vdash (\forall A. D \Rightarrow t) \preceq \sigma}$$

$$\frac{A \cap \mathrm{FV}(C, \sigma) = \varnothing \quad C \wedge D \vdash \sigma \preceq t}{C \wedge \exists A. D \vdash \sigma \preceq (\forall A. D \Rightarrow t)} \qquad \frac{C \equiv_X D \quad C \vdash \sigma \preceq \tau}{D \vdash \sigma \preceq \tau}$$

$\preceq$ は左右の α 同値で閉じる.

$$\begin{aligned} \mathrm{Inst}_\rho(t) &= \{u \mid \langle t \rangle_\rho \leq u\}, \\ \mathrm{Inst}_\rho(\forall A. D \Rightarrow t) &= \{u \mid \exists \nu =_{\overline{A}} \rho.\ \nu \models D \wedge \langle t \rangle_\nu \leq u\}, \\ \lfloor \Gamma \rfloor_\rho &= \mathrm{map}(\mathrm{Inst}_\rho, \Gamma), \\ \mathrm{GReal}(\Gamma, \rho) &\iff \forall \sigma \in \Gamma.\ \mathrm{Inst}_\rho(\sigma) \neq \varnothing, \\ \mathrm{Real}_C(\Gamma) &\iff \forall \sigma \in \Gamma.\ \exists t.\ C \vdash \sigma \preceq t. \end{aligned}$$

$$\frac{\mathrm{Real}_C(\Gamma) \quad \rho \models C}{\mathrm{GReal}(\Gamma, \rho)} \qquad \frac{C; \Gamma \vdash e : t ! \epsilon \quad \rho \models C \quad \mathrm{GReal}(\Gamma, \rho)}{\lfloor \Gamma \rfloor_\rho \vdash_g e : \langle t \rangle_\rho ! \langle \epsilon \rangle_\rho}$$

$\Gamma' = \theta\Gamma$ （意味的環境代入） $\iff$ $\forall \rho.\ \lfloor \Gamma' \rfloor_\rho = \lfloor \Gamma \rfloor_{\theta^* \rho}$.

4　固定した制約生成算法

N は簡約器：$N(D) \equiv_X D$．$\mathcal{A}(e, \Gamma, t, \epsilon; F)$ は部分関数．

$$
\begin{aligned}
f_T^F(i) &= 1 + \max(F_T \cup \{0\}) + i, & f_E^F(i) &= 1 + \max(F_E \cup \{0\}) + i,\\
P(\Gamma, t, \epsilon) &= \operatorname{supp}\Gamma \cup \operatorname{supp} t \cup (\varnothing, \mathrm{FV}_E\,\epsilon),\\
\mathrm{Check}(D, a, t, \eta, \epsilon) &= \top \wedge D \wedge (a \le t) \wedge (\eta = \epsilon),\\
\mathrm{gen}_\Gamma(D, a) &= \forall A.D \Rightarrow a, & A &= (\operatorname{supp} D \cup \operatorname{supp} a) \setminus \operatorname{supp}\Gamma.
\end{aligned}
$$

各再帰呼出しが成功したとき，以下の制約を返す．未定義な変数・操作では missing.

$$
\begin{aligned}
\mathcal{A}(i, \Gamma, t, \epsilon; F) &= \begin{cases} \mathrm{Check}(\top, a, t, \varnothing, \epsilon) & \Gamma[i] = a,\\ \mathrm{Check}(\theta D, \theta a, t, \varnothing, \epsilon) & \Gamma[i] = \forall A.D \Rightarrow a, \end{cases}\\
\theta &= [A \mapsto f^{F \cup \operatorname{supp}\top}(A)],\\
\mathcal{A}(n, \Gamma, t, \epsilon; F) &= \mathrm{Check}(\top, \mathrm{Int}, t, \varnothing, \epsilon),\\
\mathcal{A}(b, \Gamma, t, \epsilon; F) &= \mathrm{Check}(\top, \mathrm{Bool}, t, \varnothing, \epsilon).
\end{aligned}
$$

$$
\begin{gathered}
a = \alpha_{f_T^F(0)}, \quad b = \alpha_{f_T^F(1)}, \quad \eta = \mu_{f_E^F(0)},\\
F' = F \cup P(a :: \Gamma, b, \eta) \cup (\{f_T^F(0), f_T^F(1)\}, \{f_E^F(0)\}),\\
D = \mathcal{A}(e, a :: \Gamma, b, \eta; F'),\\
\boxed{\mathcal{A}(\lambda.e, \Gamma, t, \epsilon; F) = \mathrm{Check}(D, a \xrightarrow{\eta} b, t, \varnothing, \epsilon)}.
\end{gathered}
$$

$$
\begin{gathered}
a = \alpha_{f_T^F(0)}, \quad b = \alpha_{f_T^F(1)}, \quad (\eta_f, \eta_x, \eta) = (\mu_{f_E^F(0)}, \mu_{f_E^F(1)}, \mu_{f_E^F(2)}),\\
K = F \cup P(\Gamma, a, \eta_f) \cup P(\Gamma, b, \eta_x) \cup (\varnothing, \mathrm{FV}_E\,\eta),\\
D_f = \mathcal{A}(f, \Gamma, a, \eta_f; K), \quad D_x = \mathcal{A}(x, \Gamma, b, \eta_x; K \cup \operatorname{supp} D_f),\\
\boxed{\mathcal{A}(f\,x, \Gamma, t, \epsilon; F) = \top \wedge D_f \wedge D_x \wedge a \le b \xrightarrow{\eta} t \wedge \eta_f \cup \eta_x \cup \eta = \epsilon}.
\end{gathered}
$$

$$
\begin{gathered}
\mathrm{Value}(v), \quad a = \alpha_{f_T^F(0)}, \quad D_v = \mathcal{A}(v, \Gamma, a, \varnothing; F \cup P(\Gamma, a, \varnothing)),\\
\sigma = \mathrm{gen}_\Gamma(N(D_v), a), \quad A = (\operatorname{supp} N(D_v) \cup \operatorname{supp} a) \setminus \operatorname{supp}\Gamma, \quad Q = \exists A.N(D_v),\\
D_b = \mathcal{A}(e, \sigma :: \Gamma, t, \epsilon; F \cup P(\sigma :: \Gamma, t, \epsilon) \cup \operatorname{supp} Q),\\
\boxed{\mathcal{A}(\mathbf{let}\ v\ \mathbf{in}\ e, \Gamma, t, \epsilon; F) = \top \wedge Q \wedge D_b}.
\end{gathered}
$$

$$
\begin{gathered}
\neg\,\mathrm{Value}(v), \quad a = \alpha_{f_T^F(0)}, \quad (\eta_v, \eta_b) = (\mu_{f_E^F(0)}, \mu_{f_E^F(1)}),\\
K = F \cup P(\Gamma, a, \eta_v) \cup P(a :: \Gamma, t, \eta_b),\\
D_v = \mathcal{A}(v, \Gamma, a, \eta_v; K), \quad D_b = \mathcal{A}(e, a :: \Gamma, t, \eta_b; K \cup \operatorname{supp} D_v),\\
\boxed{\mathcal{A}(\mathbf{let}\ v\ \mathbf{in}\ e, \Gamma, t, \epsilon; F) = \top \wedge D_v \wedge D_b \wedge (\eta_v \cup \eta_b = \epsilon)}.
\end{gathered}
$$

5　操作の生成と射影

$$\Sigma(o) = (B, a, b), \quad \theta = [B \mapsto f^F(B)], \quad K = F \cup \mathrm{supp}(\theta a, \theta b), \quad \eta = \mu_{f_E^K(0)},$$
$$D = \mathcal{A}(u, \Gamma, \theta a, \eta; K \cup P(\Gamma, \theta a, \eta)),$$
$$\boxed{\mathcal{A}(\mathbf{perform}\ o\ u, \Gamma, t, \epsilon; F) = \mathrm{Check}(D \wedge \theta a \le \theta a, \theta b, t, \{o\} \cup \eta, \epsilon)}.$$

Handler

$$\begin{aligned}
t_i &= \alpha_{f_T^F(0)}, & t_r &= \alpha_{f_T^F(1)},\\
(\epsilon_i, \epsilon_r, \delta) &= (\mu_{f_E^F(0)}, \mu_{f_E^F(1)}, \mu_{f_E^F(2)}),\\
K &= F \cup \mathrm{supp}(t_i, t_r) \cup (\varnothing, \mathrm{FV}_E(\epsilon_i, \epsilon_r, \delta)),\\
D_i &= \mathcal{A}(e, \Gamma, t_i, \epsilon_i; K),\\
K_r &= K \cup \mathrm{supp}\, D_i, & D_r &= \mathcal{A}(e_r, t_i :: \Gamma, t_r, \epsilon_r; K_r),\\
K_o &= K_r \cup \mathrm{supp}\, D_r, & (B, a, b) &= \mathrm{rename}_{f^{K_o}}(\Sigma(o)),\\
\Gamma_o &= (b \xrightarrow{\chi} t_r) :: a :: \Gamma,\\
F_o &= K_o \cup P(\Gamma_o, t_r, \chi) \cup \mathrm{supp}(a, b) \cup B,\\
\epsilon_o &= \mu_{f_E^{F_o}(0)}, & D_o &= \mathcal{A}(e_o, \Gamma_o, t_r, \epsilon_o; F_o \cup P(\Gamma_o, t_r, \epsilon_o)),\\
J &= (\mathrm{supp}\, D_o \cup (\varnothing, \mathrm{FV}_E\, \epsilon_o)) \setminus F_o.
\end{aligned}$$

$$\begin{aligned}
H = {} & \top \wedge D_i \wedge D_r \wedge t_r \le t_r\\
& \wedge \big(\forall B. \exists J. (D_o \wedge t_r \le t_r \wedge \epsilon_o \subseteq \chi)\big)\\
& \wedge \mathrm{Split}_o(\epsilon_i, \delta) \wedge \delta \subseteq \chi \wedge \epsilon_r \subseteq \chi,\\
\mathcal{A}(\mathbf{handle}_o(e, e_r, e_o), \Gamma, t, \chi; F) = {} & \mathrm{Check}(H, t_r, t, \chi, \chi).
\end{aligned}$$

外側の射影と solver 呼出し

$$\begin{aligned}
F &= P(\Gamma, t, \epsilon) \cup \mathrm{supp}\, \top, \quad D = \mathcal{A}(e, \Gamma, t, \epsilon; F),\\
\mathcal{P}(\Gamma, e, t, \epsilon) &= \exists(\mathrm{supp}\, D \setminus F).D,\\
\mathcal{I}^{\mathrm{at}}_{N,S}(\Gamma, e, t, \epsilon) &= S(\top, \mathcal{P}(\Gamma, e, t, \epsilon)),\\
p &= f_T^{\mathrm{supp}\,\Gamma}(0), \quad q = f_E^{\mathrm{supp}\,\Gamma}(0),\\
\mathcal{I}_{N,S}(\Gamma, e) &= \mathcal{I}^{\mathrm{at}}_{N,S}(\Gamma, e, \alpha_p, \mu_q).
\end{aligned}$$

missing は後続へ伝播する．solved(R, θ) の結果は

$$(R, \theta\Gamma, \theta\alpha_p, \theta\mu_q).$$

閉環境では $\mathrm{generalize}(\Gamma, R, \theta\alpha_p, \theta\mu_q)$ を返す型スキームとする．

6 solver の契約

$$S : \mathcal{C}_X \times \mathcal{C}_X \longrightarrow \mathsf{solved}(R,\theta) \mid \mathsf{unresolved}(R) \mid \mathsf{inconsistent},$$
$$[\theta] = \bigwedge_{i \in \mathrm{dom}_T\,\theta} (\alpha_i = \theta_T(i)) \wedge \bigwedge_{j \in \mathrm{dom}_E\,\theta} (\mu_j = \theta_E(j)),$$
$$\mathrm{SolvedSubst}(\theta) \iff \begin{cases} \forall i \in \mathrm{dom}_T\,\theta.\ \mathrm{dom}\,\theta \cap \mathrm{supp}(\theta_T(i)) = \varnothing, \\ \forall j \in \mathrm{dom}_E\,\theta.\ \mathrm{dom}_E\,\theta \cap \mathrm{FV}_E(\theta_E(j)) = \varnothing. \end{cases}$$

部分 solver（S1・S2・P1 型の契約 [2]）

$$S(G,W) = \mathsf{solved}(R,\theta) \Longrightarrow \begin{cases} \mathrm{SolvedSubst}(\theta), \\ \mathrm{dom}\,\theta \cap \mathrm{FV}(G,R) = \varnothing, \\ G \wedge R \models_X \theta W, \\ G \wedge W \models_X R \wedge [\theta], \\ \mathrm{Sat}(G \wedge R). \end{cases}$$

$$S(G,W) = \mathsf{unresolved}(R) \Rightarrow G \wedge R \equiv_X G \wedge W,$$
$$S(G,W) = \mathsf{inconsistent} \Rightarrow \neg\,\mathrm{Sat}(G \wedge W).$$

完全性の追加契約

$$\boxed{\mathrm{Complete}(S) \iff \forall G, W.\ \mathrm{Sat}(G \wedge W) \Rightarrow \exists R, \theta.\ S(G,W) = \mathsf{solved}(R,\theta)}.$$

W は handler の量化制約も含む．$R = \top$ は要求しない．

補題 6.1 (解集合の保存)**.**

$$S(G,W) = \mathsf{solved}(R,\theta) \Longrightarrow G \wedge W \equiv_X G \wedge R \wedge [\theta].$$

証明.

$$\rho \models G \wedge W \Rightarrow \rho \models R \wedge [\theta],$$
$$\rho \models G \wedge R \wedge [\theta] \Rightarrow \theta^*\rho \models W \ \wedge\ \theta^*\rho = \rho \Rightarrow \rho \models W.$$

□

契約のモデル

$$S_?(G,W) = \mathsf{unresolved}(W),$$
$$S_{\mathrm{sem}}(G,W) = \begin{cases} \mathsf{solved}(W, \mathrm{id}) & \mathrm{Sat}(G \wedge W), \\ \mathsf{inconsistent} & \neg\,\mathrm{Sat}(G \wedge W). \end{cases}$$

$$\mathrm{PartialSolver}(S_?) \qquad \mathrm{PartialSolver}(S_{\mathrm{sem}}) \wedge \mathrm{Complete}(S_{\mathrm{sem}}).$$

S_{sem} は非計算的なモデル．計算的構成は後述の $\mathrm{SatDecider}(X)$ を用いる．

7　生成と合成の定理

この節の健全性では Closed(Σ) を仮定する.

定理 7.1 (固定した生成算法)**.**

$$\begin{gathered} P(\Gamma,t,\epsilon)\subseteq F,\quad \mathcal{A}(e,\Gamma,t,\epsilon;F)=D \\ \Longrightarrow\quad \top;\Gamma\vdash_N e:t!\epsilon\Rightarrow D, \\ \exists\rho.\ \lfloor\Gamma\rfloor_\rho\vdash_g e:\langle t\rangle_\rho!\langle\epsilon\rangle_\rho \\ \Longrightarrow\quad \exists D.\ \mathcal{A}(e,\Gamma,t,\epsilon;F)=D\ \wedge \\ \forall\rho.\ \lfloor\Gamma\rfloor_\rho\vdash_g e:\langle t\rangle_\rho!\langle\epsilon\rangle_\rho\Rightarrow\exists\nu=_F\rho.\ \nu\models D. \end{gathered}$$

証明. e に関する構造帰納法.

$$\begin{aligned} \lambda,\mathrm{app},\mathrm{let},\mathrm{perform}:\quad & \rho\xmapsto{\mathrm{fresh}(F)}\rho_0\xmapsto{\mathrm{IH}_1}\rho_1\xmapsto{\mathrm{IH}_2}\rho_2, \\ & \rho_{i+1}=_{F\cup\mathrm{supp}\,D_1\cup\cdots\cup\mathrm{supp}\,D_i}\rho_i; \\ \mathrm{handler}:\quad & \forall\rho_B.\ \exists\rho_J.\ \rho_J\models D_o\wedge t_r\le t_r\wedge\epsilon_o\subseteq\chi, \\ & J\cap F_o=\varnothing,\quad B\cap K_o=\varnothing,\quad \mathrm{Scope}(B,J;H,\ldots). \end{aligned}$$

□

定理 7.2 (射影後の正確性)**.**

$$\mathcal{P}(\Gamma,e,t,\epsilon)=D\wedge\mathrm{GReal}(\Gamma,\rho)\Longrightarrow\big(\rho\models D\iff\lfloor\Gamma\rfloor_\rho\vdash_g e:\langle t\rangle_\rho!\langle\epsilon\rangle_\rho\big).$$

証明. ($\Rightarrow$)：生成健全性と導出の意味解釈. ($\Leftarrow$)：$\nu=_F\rho$ を前定理で得て，$\omega=\rho[\mathrm{supp}\,D_{\mathrm{raw}}\setminus F\mapsto\nu]$ を存在証人とする. □

定理 7.3 (成功時の健全性)**.**

$$\begin{gathered} \mathcal{I}^{\mathrm{at}}_{N,S}(\Gamma,e,t,\epsilon)=\mathsf{solved}(R,\theta),\quad \rho\models R,\quad \mathrm{GReal}(\Gamma,\theta^*\rho) \\ \Longrightarrow\quad \lfloor\Gamma\rfloor_{\theta^*\rho}\vdash_g e:\langle\theta t\rangle_\rho!\langle\theta\epsilon\rangle_\rho. \end{gathered}$$

$$\Gamma'=\theta\Gamma,\quad \mathrm{Real}_\top(\Gamma),\quad \mathrm{Adm}_{\mathrm{Sat}}(\Gamma')\quad\Longrightarrow\quad \mathrm{Typing}_{\mathrm{Sat}}(R,\Gamma',e,\theta t,\theta\epsilon).$$

証明. $\rho\models R\Rightarrow\theta^*\rho\models D\Rightarrow\lfloor\Gamma\rfloor_{\theta^*\rho}\vdash_g e:\langle t\rangle_{\theta^*\rho}!\langle\epsilon\rangle_{\theta^*\rho}$. $\mathrm{Sat}(R)$ と `Typing.of_ground` により再構成する. □

定理 7.4 (solver の契約を仮定した完全性)**.**

$$\boxed{\mathrm{Complete}(S)\wedge\mathrm{Real}_C(\Gamma)\wedge\mathrm{Typing}_{\mathrm{Sat}}(C,\Gamma,e,\sigma,\epsilon)\Rightarrow\exists R,\theta.\ \mathcal{I}_{N,S}(\Gamma,e)=\mathsf{solved}(R,\theta)}.$$

証明. $\rho\models C\Rightarrow\lfloor\Gamma\rfloor_\rho\vdash_g e:u!E$. 新鮮な結果名に (u,E) を割り当てる. 生成完全性より $\mathrm{Sat}(D)$, $\mathrm{Complete}(S)$ より $S(\top,D)=\mathsf{solved}(R,\theta)$. □

8 実際の出力の主要性

$$\frac{A \cap \mathrm{FV}(\Gamma) = \varnothing \quad \mathrm{Adm}_{\mathrm{Sat}}(\exists A.C) \quad \mathrm{Typing}_{\mathrm{Sat}}(C, \Gamma, e, t, \epsilon)}{\mathrm{ComputationTyping}_{\mathrm{Sat}}(\exists A.C, \Gamma, e, \forall A.C \Rightarrow (t!\epsilon))}$$

ComputationTyping は許容される制約への強化で閉じる．

$$\begin{aligned}
\mathrm{generalize}(\Gamma, R, t, \epsilon) &= \forall A.R \Rightarrow (t!\epsilon), \\
A &= (\mathrm{supp}\, R \cup \mathrm{supp}\, t \cup (\varnothing, \mathrm{FV}_E\, \epsilon)) \setminus \mathrm{supp}\, \Gamma, \\
s &= \forall A.R \Rightarrow (t!\epsilon), \\
\mathrm{FinInst}(s; C, u, \eta) &\iff \exists \phi(\text{有限}).\ \mathrm{dom}\, \phi \subseteq A \wedge \\
&\qquad \forall \rho \models C.\ \phi^* \rho \models R \wedge \langle t \rangle_{\phi^* \rho} \leq \langle u \rangle_\rho \wedge \langle \epsilon \rangle_{\phi^* \rho} = \langle \eta \rangle_\rho.
\end{aligned}$$

定理 8.1 (開環境の有限因子)**.**

$$\begin{gathered}
\mathcal{I}^{\mathrm{at}}_{N,S}(\Gamma, e, t, \epsilon) = \mathsf{solved}(R, \theta), \qquad \Gamma' = \sigma\Gamma, \quad \mathrm{Real}_C(\Gamma'), \\
\mathrm{Typing}_{\mathrm{Sat}}(C, \Gamma', e, \sigma t, \sigma\epsilon) \\
\Longrightarrow \quad \exists \phi(\text{有限}).\ \forall \rho \models C.\ \phi^* \rho \models R \wedge \theta^*(\phi^* \rho) = \sigma^* \rho.
\end{gathered}$$

証明. $\phi = \sigma$ とする．

$$\rho \models C \Rightarrow \sigma^* \rho \models D \Rightarrow \sigma^* \rho \models R \wedge [\theta] \Rightarrow \theta^*(\sigma^* \rho) = \sigma^* \rho.$$

□

定理 8.2 (閉環境の主要計算型・主要値型)**.**

$$\begin{gathered}
\mathrm{Closed}(\Sigma), \quad \mathrm{supp}\, \Gamma = \varnothing, \quad \mathrm{Real}_\top(\Gamma), \quad \mathrm{Adm}_{\mathrm{Sat}}(\Gamma), \\
\mathcal{I}_{N,S}(\Gamma, e) = \mathsf{solved}(R, \theta), \quad p = f_T^{\mathrm{supp}\, \Gamma}(0), \quad q = f_E^{\mathrm{supp}\, \Gamma}(0), \\
s = \mathrm{generalize}(\Gamma, R, \theta\alpha_p, \theta\mu_q) \\
\Longrightarrow \quad \mathrm{ComputationTyping}_{\mathrm{Sat}}(\top, \Gamma, e, s) \wedge \\
\forall C, t, \epsilon.\ \mathrm{Typing}_{\mathrm{Sat}}(C, \Gamma, e, t, \epsilon) \Rightarrow \mathrm{FinInst}(s; C, t, \epsilon).
\end{gathered}$$

$s = \forall A.R \Rightarrow (u!\eta)$ と書く．値なら $(\top, \forall A.R \Rightarrow u)$ は主要対で，即時効果は空．

証明. 生成制約を D とする．$\mathrm{Sat}(R) \wedge R \models \theta D \Rightarrow \mathrm{Sat}(D)$．生成の主要性と実行結果の一意性から，実際の D の有限被覆を得る．

$$\begin{aligned}
\rho \models C \Rightarrow \sigma^* \rho \models D \wedge \langle \alpha_p \rangle_{\sigma^* \rho} \leq \langle t \rangle_\rho \wedge \langle \mu_q \rangle_{\sigma^* \rho} = \langle \epsilon \rangle_\rho, \\
D \equiv_X R \wedge [\theta] \Rightarrow \sigma^* \rho \models R \wedge \theta^*(\sigma^* \rho) = \sigma^* \rho.
\end{aligned}$$

σ を s の束縛名へ制限し，同じ有限因子を全ての $\rho \models C$ に用いる．値の場合は `ComputationTyping.principal_value` を適用する． □

$$\begin{aligned}
\mathcal{I}^{\mathrm{at}}_{N,S}(\Gamma, e, t, \epsilon) = \mathsf{inconsistent} &\Rightarrow \neg \exists \rho.\ \lfloor \Gamma \rfloor_\rho \vdash_g e : \langle t \rangle_\rho ! \langle \epsilon \rangle_\rho, \\
\mathcal{I}^{\mathrm{at}}_{N,S}(\Gamma, e, t, \epsilon) = \mathsf{unresolved}(R) &\Rightarrow \exists D.\ \mathcal{P}(\Gamma, e, t, \epsilon) = D \wedge R \equiv_X D.
\end{aligned}$$

9　抽象 X の完全性：HM(X) との対応

比較対象：[1, §4.4.1, 定理 14・15・系 6]．現行の型付け・生成は値制限付き．型項再構成（§4.4.2）は次節で区別する．実行時の型・効果安全性は未証明.

前提	現行パラメータ・条件
型代数・代入・有限台	`TypeDomain`, `Constraints.holds_subst`, `holds_support`
論理演算・射影	`holds_top`, `holds_and`, `holds_exists`
比較・効果	`TypeOrder`, `holds_le`, `holds_effectLe`, `holds_lacks`
同値簡約	$N(C) \equiv_X C$（`Simplifier`）
解決済み集合	$\mathcal{S} = \{C \mid \mathrm{Sat}_X(C)\}$
環境	$\mathrm{Real}_C(\Gamma)$, $\mathrm{Adm}_{\mathrm{Sat}}(\Gamma)$
操作宣言	$\mathrm{Closed}(\Sigma)$（健全性）
handler の拡張	`holds_clause`：$\forall B.\exists J.D$

$$
\begin{aligned}
\mathrm{Real}_C(\Gamma) &\iff \forall \sigma \in \Gamma.\ \exists t.\ C \vdash_i \sigma \succeq t,\\
\mathrm{Adm}_{\mathrm{Sat}}(C) &\iff \mathrm{Sat}_X(C),\\
\mathrm{Sat}_X(\exists A.C) &\iff \mathrm{Sat}_X(C),\\
\mathrm{Value}(v) &\Rightarrow \text{値 let の一般化}, \qquad \neg\,\mathrm{Value}(e) \Rightarrow \text{単相 let}.
\end{aligned}
$$

制約を保持する主要正規形

$$
\begin{aligned}
p = (C,\theta) \in \mathrm{ConstraintProblem},\quad \mathrm{Sat}_X(C)\\
\Longrightarrow\quad \mathrm{PrincipalNormal}_{\mathcal{S}}(p,p),\\
\mathrm{PrincipalNormal}_{\mathcal{S}}(p,n) \iff \mathrm{Normal}_{\mathcal{S}}(p,n) \wedge \forall m.\ \mathrm{Normal}_{\mathcal{S}}(p,m) \Rightarrow n \succeq m.
\end{aligned}
$$

`NormalForm.MoreGeneral` の反射性と `IsNormalOf` の定義による．$\mathcal{S}$ の射影閉性・主要正規形の存在に判定器は不要.

$$
\begin{aligned}
s = \forall A.R \Rightarrow (t!\eta),\quad \mathrm{valuePart}(s) = \forall A.R \Rightarrow t,\\
\mathrm{CInst}_\rho(s) = \{(u,E) \mid \exists \nu =_{\overline{A}} \rho.\ \nu \models R \wedge \langle t\rangle_\nu \le u \wedge \langle \eta\rangle_\nu = E\},\\
\mathrm{PrincipalComp}(\Gamma,e,s) \iff \mathrm{ComputationTyping}_{\mathrm{Sat}}(\top,\Gamma,e,s) \wedge\\
\qquad \forall C,u,\epsilon.\ \mathrm{Typing}_{\mathrm{Sat}}(C,\Gamma,e,u,\epsilon) \Rightarrow \forall \rho \models C.\ (\langle u\rangle_\rho, \langle \epsilon\rangle_\rho) \in \mathrm{CInst}_\rho(s),\\
\mathrm{PrincipalPair}(\Gamma,e,C,\sigma) \iff \mathrm{Typing}_{\mathrm{Sat}}(C,\Gamma,e,\sigma,\varnothing) \wedge\\
\qquad \forall D,\tau.\ \mathrm{Typing}_{\mathrm{Sat}}(D,\Gamma,e,\tau,\varnothing) \Rightarrow D \models C \wedge D \vdash_i \sigma \succeq \tau.
\end{aligned}
$$

定理 9.1 (判定器なしの生成主要性：系 6 との対応)**.**

$$
\begin{aligned}
\mathrm{supp}\,\Gamma = \varnothing,\quad \mathrm{Real}_\top(\Gamma),\quad \mathrm{Typing}_{\mathrm{Sat}}(C,\Gamma,e,\sigma,\epsilon)\\
\Longrightarrow\quad \exists D.\ \mathcal{P}(\Gamma,e,\alpha_0,\mu_0) = D \wedge \mathrm{Sat}_X(D) \wedge\\
\mathrm{PrincipalComp}(\Gamma,e,s_D) \wedge \forall C',t',\eta'.\ \mathrm{Typing}_{\mathrm{Sat}}(C',\Gamma,e,t',\eta') \Rightarrow \mathrm{FinInst}(s_D;C',t',\eta'),\\
s_D = \forall(\{\alpha_0\},\{\mu_0\}).D \Rightarrow (\alpha_0!\mu_0).
\end{aligned}
$$

`Typing.principal_computation_run`．値の主要対は `Typing.principal_value`．閉環境の系は `Typing.principal_closed_from_open` でも導く.

開環境の出力と比較

$$
\begin{aligned}
s = \mathrm{gen}(\Gamma, D, \alpha, \mu) &= \forall A.D \Rightarrow (\alpha!\mu),\\
A = (\mathrm{supp}\, D \cup \mathrm{supp}\, \alpha \cup \mathrm{supp}\, \mu) &\setminus \mathrm{supp}\, \Gamma, \qquad P_s = \exists A.D, \qquad V_s = \forall A.D \Rightarrow \alpha,\\
\mathrm{Cover}(\Gamma, e, s) \iff \forall C, \sigma, \epsilon.\ &\left(\mathrm{Typing}_{\mathrm{Sat}}(C, \Gamma, e, \sigma, \epsilon) \wedge \mathrm{Real}_C(\Gamma)\right)\\
&\Rightarrow C \models P_s \ \wedge\ C \vdash_i V_s \succeq \sigma,\\
\mathrm{FinCover}(\Gamma, e, s) \iff \forall C, u, \epsilon.\ &\left(\mathrm{Typing}_{\mathrm{Sat}}(C, \Gamma, e, u, \epsilon) \wedge \mathrm{Real}_C(\Gamma)\right)\\
&\Rightarrow \mathrm{FinInst}(s; C, u, \epsilon).
\end{aligned}
$$

定理 9.2 (開環境の生成完全性・一般性：定理 15 との対応)**.**

$$
\begin{aligned}
&\mathrm{Typing}_{\mathrm{Sat}}(C, \Gamma, e, \sigma, \epsilon), \quad \mathrm{Real}_C(\Gamma), \quad \alpha, \mu \notin \mathrm{supp}\, \Gamma\\
&\Longrightarrow \exists D.\ \mathcal{P}(\Gamma, e, \alpha, \mu) = D \wedge \mathrm{Sat}_X(D) \ \wedge\\
&\mathrm{Typing}_{\mathrm{Sat}}(D, \Gamma, e, \alpha, \mu) \ \wedge\\
&\mathrm{ComputationTyping}_{\mathrm{Sat}}(P_s, \Gamma, e, s) \ \wedge \mathrm{Cover}(\Gamma, e, s) \wedge \mathrm{FinCover}(\Gamma, e, s).
\end{aligned}
$$

$$
C \models P_s, \qquad C \vdash_i V_s \succeq \sigma, \qquad \sigma = u \Rightarrow \mathrm{FinInst}(s; C, u, \epsilon).
$$

`Typing.principal_computation_open_at`, `Typing.inference_complete_scheme`.

証明.

$$
\begin{aligned}
\rho \models C \Rightarrow \exists \nu =_{\overline{\{\alpha,\mu\}}} \rho.\ \nu \models D \Rightarrow \rho \models P_s,\\
\theta_{u,\eta} = [\alpha \mapsto u, \mu \mapsto \eta], \qquad \lfloor \Gamma \rfloor_{\theta^*_{u,\eta}\rho} = \lfloor \Gamma \rfloor_\rho,\\
\mathrm{Real}_{C'}(\Gamma),\ C', \Gamma \vdash e : u!\eta \Rightarrow \forall \rho \models C'.\ \theta^*_{u,\eta}\rho \models D \Rightarrow \mathrm{FinInst}(s; C', u, \eta).
\end{aligned}
$$

qualified 型：新鮮化・単相化・∀ 右規則. □

定理 9.3 (開環境の値の主要対)**.**

$$
\begin{aligned}
&\mathrm{Typing}_{\mathrm{Sat}}(C, \Gamma, e, \sigma, \epsilon), \quad \mathrm{Real}_\top(\Gamma), \quad \mathrm{Value}(e)\\
&\Longrightarrow \exists s.\ \mathrm{PrincipalPair}(\Gamma, e, P_s, V_s).
\end{aligned}
$$

`Typing.principal_value_open`.

外側条件の保持

$$
\begin{aligned}
C = (\alpha_0 = \mathrm{Int}), \quad \Gamma = [x : \forall(\{\alpha_1\}, \{\mu_1\}).C \Rightarrow \alpha_1], \quad e = x,\\
\mathrm{Real}_C(\Gamma), \quad \neg\, \mathrm{Real}_\top(\Gamma), \quad P_s \equiv_X C, \quad \top \not\models P_s.
\end{aligned}
$$

`OpenExamples.qualified_outer_condition`.

抽象的な充足可能性判定器

$$\boxed{\mathrm{SatDecider}(X) = \{d : \mathcal{C}_X \to \mathbb{B} \mid \forall C.\ d(C) = \mathsf{true} \iff \mathrm{Sat}_X(C)\}.}$$

d は停止する計算可能な実装を与える．型クラス・単一化方式は固定しない．

$$S_d(G, W) = \begin{cases} \mathsf{solved}(W, \mathrm{id}) & d(G \wedge W) = \mathsf{true}, \\ \mathsf{inconsistent} & d(G \wedge W) = \mathsf{false}. \end{cases}$$

定理 9.4 (判定器からの完全な solver)**.**

$$\begin{gathered} d \in \mathrm{SatDecider}(X) \Rightarrow \mathrm{PartialSolver}(S_d) \wedge \mathrm{Complete}(S_d), \\ S_d(G, W) = \mathsf{solved}(R, \theta) \iff \mathrm{Sat}_X(G \wedge W) \wedge (R, \theta) = (W, \mathrm{id}), \\ S_d(G, W) = \mathsf{inconsistent} \iff \neg\, \mathrm{Sat}_X(G \wedge W), \\ \forall G, W, R.\ S_d(G, W) \neq \mathsf{unresolved}(R). \end{gathered}$$

証明. $d(G \wedge W) = \mathsf{true} \iff \mathrm{Sat}_X(G \wedge W)$ と恒等代入の契約による． □

定理 9.5 (合成した完全性・主要性)**.**

$$\begin{gathered} d \in \mathrm{SatDecider}(X), \quad \mathrm{Real}_C(\Gamma), \quad \mathrm{Typing}_{\mathrm{Sat}}(C, \Gamma, e, \sigma, \epsilon) \\ \Longrightarrow \quad \exists n.\ \mathcal{I}_{N,S_d}(\Gamma, e) = \mathsf{solved}(n) \wedge \\ \mathrm{ComputationTyping}_{\mathrm{Sat}}(P_s, \Gamma, e, s) \ \wedge \mathrm{Cover}(\Gamma, e, s) \wedge \mathrm{FinCover}(\Gamma, e, s), \\ s = \mathrm{inferredScheme}(\Gamma, n). \end{gathered}$$

$$\begin{gathered} \mathrm{Closed}(\Sigma), \quad \mathrm{supp}\,\Gamma = \varnothing, \quad \mathrm{Real}_\top(\Gamma) \\ \Longrightarrow \quad \mathrm{PrincipalComp}(\Gamma, e, s) \wedge \\ \big(\forall C', t', \eta'.\ \mathrm{Typing}_{\mathrm{Sat}}(C', \Gamma, e, t', \eta') \Rightarrow \mathrm{FinInst}(s; C', t', \eta')\big) \wedge \\ \big(\mathrm{Value}(e) \Rightarrow \mathrm{PrincipalPair}(\Gamma, e, \top, \mathrm{valuePart}(s))\big). \end{gathered}$$

`SatDecider.infer_principal_open`, `infer_complete_scheme`.

$$\mathrm{Real}_\top(\Gamma) \wedge \mathrm{Value}(e) \Rightarrow \mathrm{PrincipalPair}(\Gamma, e, P_s, V_s).$$

`SatDecider.infer_principal_value_open`.

$$\mathrm{Typing} \xRightarrow{\text{生成完全性}} \mathrm{Sat}_X(D) \xRightarrow{d} \mathsf{solved}(D, \mathrm{id}) \xRightarrow{\text{生成・solver の契約}} \text{健全性・主要性.}$$

健全性（定理 14 との対応）：$\mathrm{Closed}(\Sigma), \mathrm{Adm}_{\mathrm{Sat}}(\Gamma)$ の下で

$$\mathcal{I}^{\mathrm{at}}_{N,S_d}(\Gamma, e, t, \epsilon) = \mathsf{solved}(R, \theta) \Rightarrow \mathrm{Typing}_{\mathrm{Sat}}(R, \Gamma, e, t, \epsilon).$$

`SatDecider.inferAt_sound_of_admissible`．実現可能性は不要．

handler の条件

$$\rho \models \forall B.\exists J.D \iff \forall \nu =_{\overline{B}} \rho.\ \exists \omega =_{\overline{J}} \nu.\ \omega \models D.$$

生成定理は handler を含む．d の判定域もこの量化制約を含む．基底断片のみの判定器から全 X の `SatDecider` は得られない．handler 固有の不完全性・判定不能性は主張しない．

10　型項再構成：有限範囲の正規化

$$
\begin{aligned}
n = (R, \psi), \qquad & [n] = R \wedge [\psi], \\
\mathrm{Normal}_S(D; n) &\iff S(R) \wedge \mathrm{SolvedSubst}(\psi) \wedge \psi R \equiv R \wedge [n] \models D, \\
n \preceq_U m &\iff \exists \phi (\text{有限代入}).\ \forall \rho.\ \psi^*(\phi^* \rho) =_U \theta^* \rho \ \wedge\ Q \models \phi R, \quad m = (Q, \theta), \\
\mathrm{Principal}^U_S(D; n) &\iff \mathrm{Normal}_S(D; n) \wedge \forall m.\ \mathrm{Normal}_S(D; m) \Rightarrow n \preceq_U m.
\end{aligned}
$$

比較因子 ϕ の冪等性は要求しない．新しい出力パラメータは入力 U と区別する．

$$
\begin{gathered}
\mathrm{FV}(D) \subseteq U, \quad S \subseteq \mathrm{Sat}_X, \\
\mathcal{N}(U, D) = \mathsf{some}(n) \Rightarrow \mathrm{Principal}^U_S(D; n), \\
\mathcal{N}(U, D) = \mathsf{none} \iff \neg \exists n. \mathrm{Normal}_S(D; n).
\end{gathered}
$$

`ScopedPrincipalNormalizer`．存在条件 `PrincipalNormalForms` と計算可能な $\mathcal{N}$ の実装を区別する．

解形式の閉性（`SolvedClosure`）

$$
\begin{gathered}
\mathrm{Adm}_S(C) \Rightarrow \mathrm{Adm}_S(pC) \quad (p : \text{有限置換}), \\
\mathrm{Adm}_S(C) \wedge \mathrm{Adm}_S(D) \wedge \mathrm{Sat}(C \wedge D) \Rightarrow \mathrm{Adm}_S(C \wedge D), \\
\mathrm{Adm}_S(C) \Rightarrow \mathrm{Adm}_S(\exists A.C).
\end{gathered}
$$

主要正規形の契約からこれらの閉性は導出していない．$S = \mathrm{Sat}_X$ は全条件を満たす．

定理 10.1 (新鮮化した比較正規形)**.** Adm_S の改名閉性，$\mathrm{FV}(D) \subseteq U$ の下で，

$$
\begin{aligned}
& \mathrm{Adm}_S(C), \quad C \models \sigma D \\
\Longrightarrow \quad & \exists n, \phi.\ \mathcal{N}(U, D) = \mathsf{some}(n) \ \wedge \\
& \forall \rho.\ n^*_\theta(\phi^* \rho) =_U \sigma^* \rho, \qquad C \models \phi n_R.
\end{aligned}
$$

`ScopedPrincipalNormalizer.complete_of_substitution`．$\mathrm{SolvedSubst}(\sigma)$，$\sigma C \equiv C$，ground 値の型項表現は不要．

捕獲回避代入

$$
\sigma(\forall A.C \Rightarrow t) = \forall f(A).\eta C \Rightarrow \eta t, \qquad \eta(i) = \begin{cases} f(i) & i \in A, \\ \sigma(i) & i \notin A, \end{cases}
$$

$$
\begin{gathered}
f(A) \cap (\mathrm{supp}\, C \cup \mathrm{supp}\, t \cup \mathrm{dom}\, \sigma \cup \mathrm{range}(\sigma) \cup A) = \varnothing, \\
\lfloor \Gamma[\sigma] \rfloor_\rho = \lfloor \Gamma \rfloor_{\sigma^* \rho}, \qquad C \vdash_i s \succeq t \Rightarrow \sigma C \vdash_i s[\sigma] \succeq \sigma t, \\
\mathrm{Real}_C(\Gamma) \Rightarrow \mathrm{Real}_{\sigma C}(\Gamma[\sigma]).
\end{gathered}
$$

比較の右辺 t は単相型．任意の qualified 型への代入保存は未証明．

再構成算法と証明範囲

$$\mathcal{T}^{\mathrm{at}}_{N,\mathcal{N}}(\Gamma, e, t, \epsilon): \quad D \leftarrow \mathcal{P}_N(\Gamma, e, t, \epsilon),$$
$$U = \mathrm{Protected}(\Gamma, t, \epsilon) \cup \mathrm{supp}\, D,$$
$$\mathcal{T}^{\mathrm{at}}_{N,\mathcal{N}} = \begin{cases} \mathsf{reconstructed}(R, \psi) & \mathcal{N}(U, D) = \mathsf{some}(R, \psi), \\ \mathsf{unrepresented}(D) & \mathcal{N}(U, D) = \mathsf{none}. \end{cases}$$

`inferTerm` は Γ に新鮮な型・効果変数を期待値に使う．unrepresented は $\neg\,\mathrm{Sat}(D)$ を意味しない．

定理 10.2 (制約形式での再構成健全性)**.**

$$\mathrm{Closed}(\Sigma), \quad \mathrm{Real}_{\top}(\Gamma), \quad \mathcal{T}^{\mathrm{at}}(\Gamma, e, t, \epsilon) = \mathsf{reconstructed}(R, \psi)$$
$$\Longrightarrow \quad \mathrm{Typing}_{\mathrm{Sat}}(R, \Gamma[\psi], e, \psi t, \psi\epsilon).$$

`inferTermAt_sound_constraint`．結論は Typing_S ではない．

定理 10.3 (単相判断の完全性と有限因子)**.** Adm_S の改名閉性の下で，

$$\mathrm{Typing}_S(C, \Gamma[\sigma], e, \sigma t, \sigma\epsilon), \quad \mathrm{Real}_C(\Gamma[\sigma])$$
$$\Longrightarrow \quad \exists R, \psi, \phi.\ \mathcal{T}^{\mathrm{at}}(\Gamma, e, t, \epsilon) = \mathsf{reconstructed}(R, \psi) \ \wedge\ C \models \phi R,$$
$$\forall \rho. \quad \lfloor \Gamma[\psi] \rfloor_{\phi^* \rho} = \lfloor \Gamma[\sigma] \rfloor_{\rho}, \quad \langle \psi t \rangle_{\phi^* \rho} = \langle \sigma t \rangle_{\rho}, \quad \langle \psi \epsilon \rangle_{\phi^* \rho} = \langle \sigma \epsilon \rangle_{\rho}.$$

`inferTermAt_complete_mono`, `reconstruction_factor`.

定理 10.4 (qualified 判断からの受理)**.** $\mathrm{SolvedClosure}(S)$ の下で，

$$\mathrm{Typing}_S(C, \Gamma[\sigma], e, s, \epsilon), \quad \mathrm{Real}_C(\Gamma[\sigma])$$
$$\Longrightarrow \quad \exists n.\ \mathcal{T}(\Gamma, e) = \mathsf{reconstructed}(n).$$

`inferTerm_complete_acceptance`．対象スキームの $\mathrm{Real}_C(s)$ は不要．

学位論文との対応	状態
定理 7：内部導出の $\mathrm{Sat}_X \to S$ 変換	未証明
定理 16：Typing_S での出力健全性	未証明
定理 17：受理・単相判断の有限因子	上記の閉性条件下で証明済み
定理 17：同じ因子による qualified 型・外側条件の比較	未証明
系 7：同じ S 内での主要型	未証明

handler の $\forall B.\exists J$ を含む生成は上記の受理定理の対象．内部導出の変換と量化・非脱出の保存は未証明．

11　具体化：正則な型クラス制約

$$K = \{k_1, \dots, k_n\}, \quad \mathrm{arity} : K \to \mathbb{N}, \qquad \mathcal{C} = \{C_1, \dots, C_m\},$$
$$t ::= \alpha \mid b \mid t \xrightarrow{\epsilon} t \mid k(t_1, \dots, t_{\mathrm{arity}(k)}), \qquad V = \{t \mid \mathrm{FV}(t) = \varnothing\}, \quad \leq = =,$$
$$D ::= \top \mid \bot \mid D \wedge D \mid t = u \mid C(t) \mid \epsilon \subseteq \eta \mid o \notin \epsilon \mid \exists A.D \mid \forall B. \exists J.D \mid \theta D.$$

θD は遅延代入：$\rho \models \theta D \iff \theta^* \rho \models D$.

$$\mathcal{I}_b \subseteq \mathcal{C} \times \mathrm{Base}, \qquad \mathcal{I}(C,k) \in \{\bot\} \cup \prod_{j < \mathrm{arity}(k)} \mathcal{P}(\mathcal{C}),$$
$$\mathcal{I}(C,k) = (P_j)_j \quad \equiv \quad \frac{\bigwedge_j \bigwedge_{D \in P_j} D(\alpha_j)}{C(k(\vec{\alpha}))}.$$

同一 head は一つ．前提は head の真部分型のみ．矢印型のインスタンスは置かない．

$$\mathrm{covers}(P,Q) \iff Q \subseteq \bigcup_{C \in P} (\{C\} \cup \mathrm{sup}(C)),$$
$$\mathrm{WF}(\mathcal{I}) \iff \big(D \in \mathrm{sup}(C) \Rightarrow D < C \wedge \mathrm{sup}(D) \subseteq \mathrm{sup}(C)\big) \wedge$$
$$\big((C,b) \in \mathcal{I}_b \Rightarrow \forall D \in \mathrm{sup}(C).\ (D,b) \in \mathcal{I}_b\big) \wedge$$
$$\big(\mathcal{I}(C,k) = (P_j)_j \Rightarrow \forall D \in \mathrm{sup}(C).\ \exists (Q_j)_j.\ \mathcal{I}(D,k) = (Q_j)_j \wedge \forall j.\, \mathrm{covers}(P_j, Q_j)\big).$$
$$\mathrm{check}(\Delta) = \mathsf{some}(\mathcal{I}, \pi) \iff \mathrm{NoDuplicateHeads}(\Delta) \wedge \mathrm{WF}(\mathcal{I}), \quad \mathcal{I} = \mathrm{environment}(\Delta).$$

有限状態による存在判定

$$q(b) = \{C \mid (C,b) \in \mathcal{I}_b\}, \qquad q(u \xrightarrow{E} v) = \varnothing,$$
$$\delta_k(q_1, \dots, q_r) = \{C \mid \mathcal{I}(C,k) = (P_j)_j \wedge \forall j.\ P_j \subseteq q_j\},$$
$$q(k(\vec{u})) = \delta_k(q(u_1), \dots, q(u_r)),$$
$$R_0 = \{\varnothing\} \cup \{q(b) \mid (C,b) \in \mathcal{I}_b\},$$
$$R_{i+1} = R_i \cup \{\delta_k(\vec{q}) \mid k \in K,\ \vec{q} \in R_i^{\mathrm{arity}(k)}\}.$$

定理 11.1.

$$\mathrm{WF}(\mathcal{I}) \Rightarrow (\mathcal{I} \vdash C(u) \iff C \in q(u)),$$
$$R = \mathrm{lfp}(R_i), \qquad q \in R \iff \exists u \in V.\ q(u) = q,$$
$$\mathrm{inhabited}(P) = \mathsf{true} \iff \exists q \in R.\ P \subseteq q \iff \exists u.\ \bigwedge_{C \in P} C \in q(u).$$

証明．構造帰納法．飽和の測度は $2^m - |R_i|$．同一変数の述語は P に集約する．　□

12　効果判定・存在射影・実行結果

効果原子の連言を F，出現する名前・ラベルを M, L とする．

$$
\begin{aligned}
Q_0 &= M \times L, & \rho_Q(\mu) &= \{o \mid (\mu, o) \in Q\},\\
\mathrm{allow}_{\epsilon \subseteq \eta}(Q, \mu, o) &\iff \mu \in \mathrm{FV}_E(\epsilon) \Rightarrow o \in \langle \eta \rangle_{\rho_Q},\\
\mathrm{allow}_{p \notin \epsilon}(Q, \mu, o) &\iff \mu \in \mathrm{FV}_E(\epsilon) \Rightarrow o \neq p,\\
Q_{i+1} &= \{(\mu, o) \in Q_i \mid \forall f \in F.\, \mathrm{allow}_f(Q_i, \mu, o)\},\\
Q_* &= \mathrm{gfp}(Q_i), & \mathrm{effectSat}(F) &= [\rho_{Q_*} \models F].
\end{aligned}
$$

定理 12.1.

$$\mathrm{effectSat}(F) = \mathsf{true} \iff \exists \rho_E.\, \rho_E \models F.$$

証明. $\rho \models F \Rightarrow (\rho \cap L) \models F$. Q_0 内の任意のモデルは各 Q_i 以下．固定点では変数由来の包含・不在が成立し，残るリテラル条件はモデルの存在から従う．停止測度は $|Q_i|$. □

射影

$$
\begin{gathered}
D \models \alpha = t, \quad \alpha \notin \mathrm{FV}_T(t), \quad \alpha \in A_T, \quad \theta = [\alpha \mapsto t] \quad \Longrightarrow \quad \exists A.D \equiv \exists (A \setminus \{\alpha\}).\theta D,\\
D \models \mu = \epsilon, \quad \mu \notin \mathrm{FV}_E(\epsilon), \quad \mu \in A_E \quad \Longrightarrow \quad \exists A.D \equiv \exists (A \setminus \{\mu\}).[\mu \mapsto \epsilon] D,\\
\mathrm{supp}\, D \subseteq A \quad \Longrightarrow \quad \rho \models \exists A.D \iff \mathrm{Sat}(D),\\
N(\forall B. \exists J. D) = \mathrm{smartClause}(B, \varnothing, \mathrm{project}(J, N(D))),\\
\boxed{N(D) \equiv D}.
\end{gathered}
$$

型・効果の束縛をまたぐ未処理の代入は suspension として残す．

$$\alpha \in \mathrm{FV}_T(t), \quad t \neq \alpha \;\Rightarrow\; |\rho_T(\alpha)| < |\langle t \rangle_\rho| \;\Rightarrow\; \neg\, \mathrm{Sat}(\alpha = t).$$

具体的 solver S_{tc}

$$
\begin{gathered}
S_{\mathrm{tc}}(G, W) = \mathsf{solved}(R, \theta) \Rightarrow \mathrm{SolvedSubst}(\theta) \wedge \theta G \equiv G \wedge \theta R \equiv R \wedge \mathrm{Sat}(G \wedge R),\\
G \wedge R \models \theta W, \qquad G \wedge W \models R \wedge [\theta],\\
S_{\mathrm{tc}}(G, W) = \mathsf{unresolved}(R) \Rightarrow G \wedge R \equiv G \wedge W,\\
S_{\mathrm{tc}}(G, W) = \mathsf{inconsistent} \Rightarrow \neg\, \mathrm{Sat}(G \wedge W).
\end{gathered}
$$

S_{tc} は有限探索・構造再帰・上記の停止測度で計算する．意味的な充足可能性 oracle は用いない．健全性・主要性は前節の定理を $X_{\mathrm{tc}}, N, S_{\mathrm{tc}}$ に適用する（同じ環境・宣言の前提）.

13　完全性の証明範囲と具体的検証

$$F ::= \top \mid F \wedge F \mid C(\alpha) \mid \epsilon \subseteq \eta \mid o \notin \epsilon,$$
$$\mathcal{D} ::= F \mid \exists A.\mathcal{D}.$$

simplify は構造的簡約，N は存在射影を含む正規化.

定理 13.1 (具体的 solver の部分体系での完全性)**.**

$$\mathrm{simplify}(G \wedge N(W)) \in \mathcal{D} \wedge \mathrm{Sat}(G \wedge W) \quad \Longrightarrow \quad \exists R, \theta.\ S_{\mathrm{tc}}(G, W) = \mathsf{solved}(R, \theta).$$

型等式の逐次消去も実装するが，全ての handler-free 生成制約が判定域へ到達する証明は未完了．全体系の Complete(S_{tc})，handler 固有の不完全性は主張しない.

$$\mathrm{Eq(Int)},\ \mathrm{Eq(Bool)},\ \mathrm{Ord(Int)},\quad \mathrm{Ord}(\alpha) \Rightarrow \mathrm{Eq}(\alpha),$$
$$C(\alpha) \Rightarrow C(\mathrm{List}\,\alpha), \qquad C(\alpha) \wedge C(\beta) \Rightarrow C(\mathrm{Pair}(\alpha, \beta)), \quad C \in \{\mathrm{Eq}, \mathrm{Ord}\}.$$

入力（$G = \top$）	実行結果
$\mathrm{Eq(List\,Bool)}$	solved
$\mathrm{Ord(List\,Bool)}$	inconsistent
$\mathrm{Eq}(\alpha)$	$\mathsf{solved}(\mathrm{Eq}(\alpha), \mathrm{id})$
$\exists\alpha.\,\mathrm{Eq}(\alpha)$	$\mathsf{solved}(\top, \mathrm{id})$
$\alpha = \mathrm{Int}$	$\mathsf{solved}(\top, [\alpha \mapsto \mathrm{Int}])$
$\alpha = \mathrm{List}\,\alpha$	inconsistent
$\forall\alpha.\exists\beta.\alpha = \beta$	solved
$\exists\beta.\forall\alpha.\alpha = \beta$	unresolved

最後の式は偽であり，未解決は検出能力の制限を表す．量化順序の交換は行わない.

$$\Sigma(o) = \forall\alpha.\alpha \to \alpha, \qquad e_h = \mathbf{handle}_o(\mathbf{perform}\ o\ 0_{\mathrm{int}},\ 0,\ 0\ 1),$$
$$[] \vdash_g e_h : \mathrm{Int}!\varnothing, \qquad \mathcal{I}_{N, S_{\mathrm{tc}}}([], e_h) = \mathsf{solved}(R, \theta).$$

後者と多相 let・未使用の Eq 束縛・perform は実行検証．前者は Lean の型付け証明．**対応表**（名前空間 `EHMX.TypeClasses`）

保証	Lean
宣言検査・クラス意味論	`Declarations.check_iff`, `holds_iff`
共同存在判定	`reachable_iff`, `inhabited_iff`
効果の完全な判定	`propagationSat_iff`
平坦な制約の完全な判定	`flatSat_iff`
型・効果の存在消去	`eliminate_bound`, `eliminate_effect`
同値正規化	`normalize_correct`
具体的 solver の契約	`solver`
具体的推論の健全性・主要性	`inferClasses_sound`, `inferClasses_principal`
上記 $\mathcal{D}$ の完全性	`solver_complete_fragment`
具体的 handler の型付け	`Examples.resumeIdentity_ground`
実行検証	`Examples.regressions`

14　抽象定理の Lean 対応

対応表（名前空間 `EHMX.HMX`）

数式・保証	Lean
$\mathcal{A}, \mathcal{P}, \mathcal{I}^{\mathrm{at}}, \mathcal{I}$	`generate`, `generateProjected`, `inferAt`, `infer`
部分・完全 solver	`PartialSolver`, `PartialSolver.Complete`
契約の非計算的モデル	`PartialSolver.semantic_complete`
充足可能性判定器	`SatDecider`
判定器から完全な solver	`SatDecider.solver_complete`
充足可能性と成功の同値	`SatDecider.solver_solved_iff`
不充足・未解決の排除	`SatDecider.solver_inconsistent_iff`, `SatDecider.solver_ne_unresolved`
射影閉性・主要正規形	`Constraints.satisfiable_exists_iff`, `Constraints.PrincipalNormalizer.satisfiable`
固定した生成の健全性	`generate_sound`
固定した生成の完全性	`Inference.groundComplete`
判定器なしの主要型の存在	`Typing.principal_computation_run`, `Typing.principal_value`
射影後の正確性	`generateProjected_ground_iff`
開環境の成功時健全性	`inferAt_ground_sound`, `inferAt_sound`
一般 solver の開環境有限因子	`infer_principal_open`
実際の結果の主要性	`infer_principal_closed`
期待型なしの条件付き完全性	`infer_complete`
判定器からの合成完全性	`SatDecider.infer_complete`
判定器からの合成健全性	`SatDecider.infer_sound_of_admissible`
開環境の生成完全性	`Typing.inference_complete_scheme`
開環境の生成主要性	`Typing.principal_computation_open_at`
qualified 型への比較	`Typing.instance_of_finite_realizable`
開環境の値の主要対	`Typing.principal_value_open`
閉環境への特殊化	`Typing.principal_closed_from_open`
開環境の合成完全性	`SatDecider.infer_complete_scheme`
開環境の合成主要性	`SatDecider.infer_principal_open`
開環境の値の主要出力	`SatDecider.infer_principal_value_open`
外側条件の保持	`OpenExamples.qualified_outer_condition`
開環境の handler	`OpenExamples.open_nested_handler_complete`
型付け可能なら主要出力	`SatDecider.infer_principal`
推論での未解決の排除	`SatDecider.infer_ne_unresolved`
不整合・未解決	`inferAt_inconsistent`, `inferAt_unresolved`
handler の具体例	`AlgorithmExamples.resumeIdentity_ground`, `resumeIdentity_generated`, `resumeIdentity_complete`, `resumeIdentity_deferred`
判定器を仮定した適用例	`AlgorithmExamples.polymorphicLet_principal_of_decider`, `resumeIdentity_principal_of_decider`, `nestedHandler_principal_of_decider`

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